Networking Fundamentals 101 – Week 2 Practical

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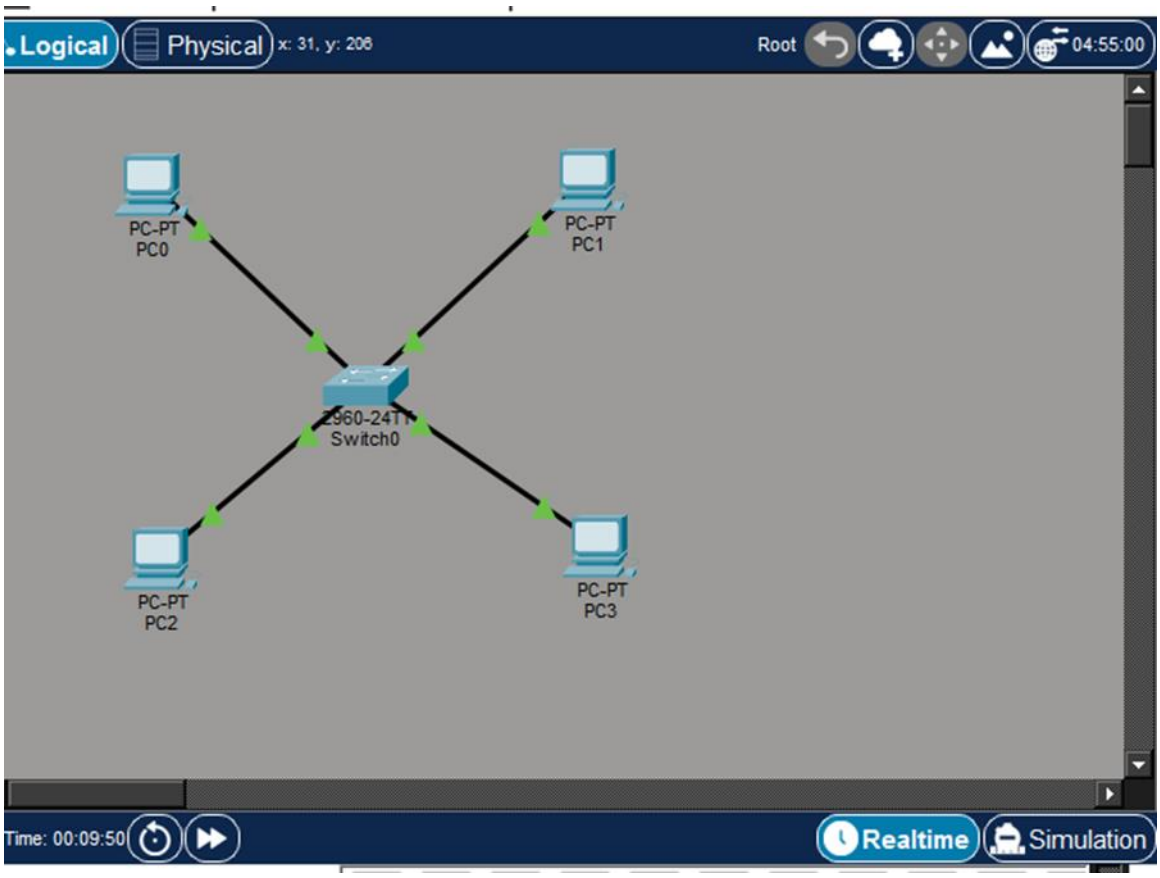
Task 1 — Star Topology

This section focuses on the design and analysis of different network topologies, specifically the star and partial mesh topologies. The practical explores how devices are interconnected, how data flows within each topology, and how network performance is affected by link status, speed, and redundancy. Using Packet Tracer, both topologies were implemented and tested to evaluate connectivity, reliability, and fault tolerance in different network scenarios.

1.2 Link Status Table



Verify link Status Table

Port	Connected Device	Status	Duplex Speed
Fa0/1 PC0	Connected	a-full	a-100
Fa0/2 PC1	connected	a-full	a-100
Fa0/3 PC2	connected	a-full	a-100
Fa0/4 PC3	Connected	a-full	a-100

```
Switch#show interfaces fastEthernet 0/1
FastEthernet0/1 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.4275.e701 (bia 0001.4275.e701)
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
     2357 packets output, 263570 bytes, 0 underruns
```

Ping Test Results

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=6ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=12ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time=10ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

```
C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time=10ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Fault Tolerance Analysis

Q1: What happens to communication between the PCs when the switch is removed? Why?

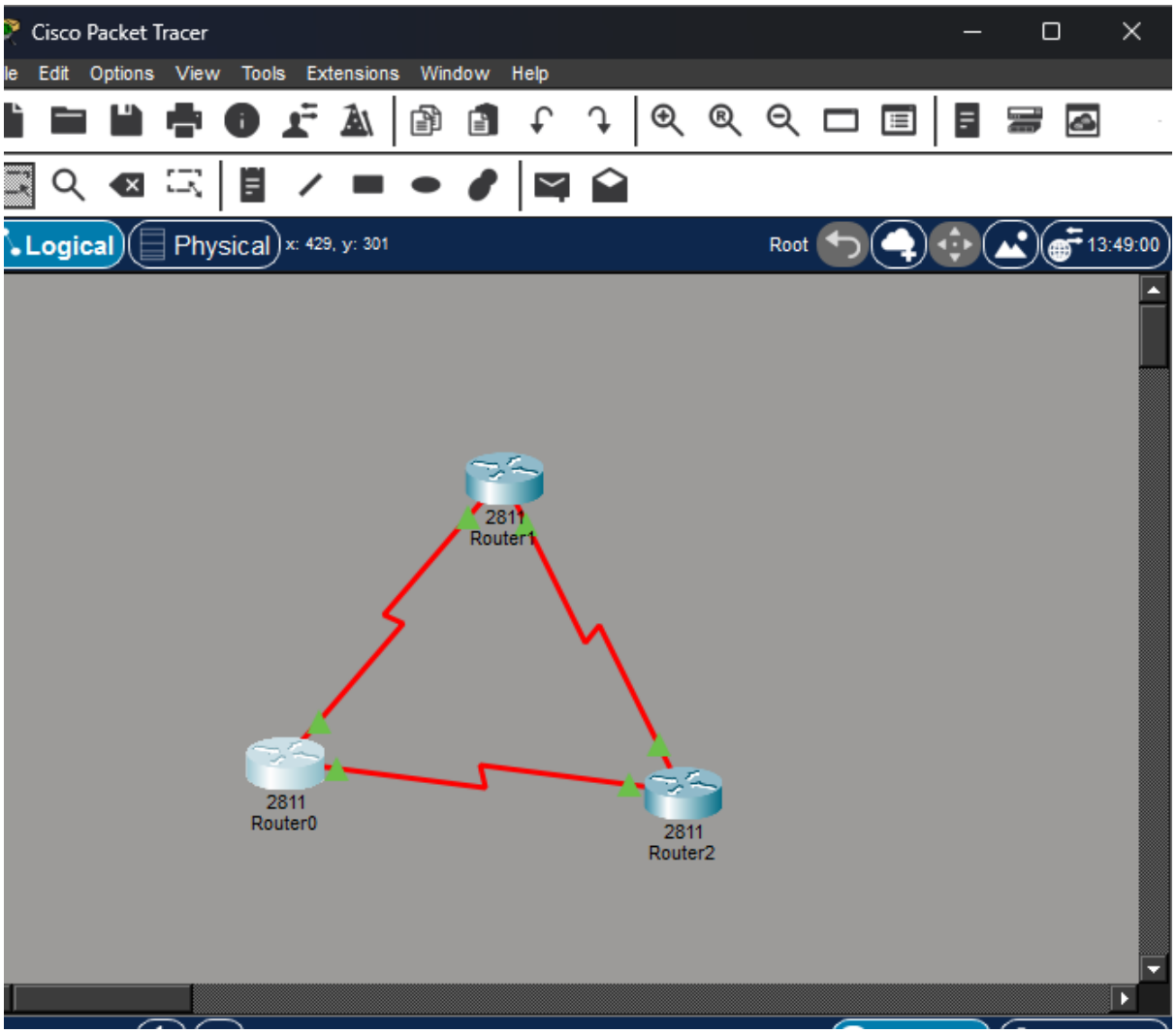
When the switch is removed, the PCs cannot communicate with each other because the switch is the central device that connects all the computers in a star topology. Without the switch there is no path for data to travel between the PCs.

Q2: Is a star topology fault tolerant? Justify your answer.

No. A star topology is not fault tolerant because if the central switch fails, the entire network stops working. All devices lose connectivity since they rely on the switch for communication.

Task 2 — Partial Mesh Topology

2.1 Build the Partial Mesh Network



2.2 Verify Interface Status

Router	Interface	IP Address	Line Status	Protocol
Router 0	Se0/0/0	10.0.0.1	up	up
Router 0	Se0/0/1	10.0.1.1	up	up
Router 1	Se0/0/0	10.0.0.2	up	up
Router 1	Se0/0/1	10.0.2.1	up	up
Router 2	Se0/0/0	10.0.1.2	up	up
Router 2	Se0/0/1	10.0.2.2	up	up

Router0

Physical Config CLI Attributes

```
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address
% Incomplete command.
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show the ip interface brief
                ^
% Invalid input detected at '^' marker.

Router# do show ip interface brief
          ^
% Invalid input detected at '^' marker.

Router# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            unassigned      YES manual up                    up
Serial0/0/1            10.0.0.1        YES manual up                    up
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router2

Physical Config CLI Attributes

```
Press RETURN to get started!


Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/1
Router(config-if)#ip address 10.0.2.1 255.255.255.252
Router(config-if)# clock rate 64000
This command applies only to DCE interfaces
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.1.2        YES manual up                    up
Serial0/0/1            10.0.2.1        YES manual up                    up
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router0

Physical Config CLI Attributes

```
Press RETURN to get started!


Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            10.0.1.1        YES manual administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router0

Physical Config CLI Attributes

```
Press RETURN to get started!


Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            10.0.1.1        YES manual administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

2.3 Connectivity and Redundancy Test

Source	Destination IP	Before Shutdown	After Shutdown
Router 0	10.0.0.2 (R1)	Success	functional
Router 0	10.0.1.2 (R2)	fail	Link down
Router 1	10.0.2.2(R2)	fail	Destination unreachable

```
Router>enable
Router# ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/9/26 ms

Router# ping 10.0.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router# ping 10.0.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds:
U.U.U
Success rate is 0 percent (0/5)

Router# ping 10.0.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds:
U.U.U
Success rate is 0 percent (0/5)
```

2.4 Redundancy Analysis

Q3: When you shut down Se0/0/0 on Router0, could Router0 still reach Router1? How?

After shutting down the Se0/0/0 interface, Router 0 cannot reach Router 1 because the Unreachable result proves there is no backup path available.

This shows that the network has no redundancy meaning once the main connection is closed all communication between the routers stops.

Q4: How does the partial mesh topology improve fault tolerance compared to the star topology?

A partial mesh topology improves fault tolerance by providing multiple backup paths so if one link fails data can be rerouted through another connection. Unlike a star topology it removes the single point of failure where a central hub going down would crash the entire network.

Q5: What is one disadvantage of using a full mesh instead of a partial mesh in a large network?

Extremely high cost and complexity of installing and managing so many physical cables and ports.

Conclusion

The practical demonstrated how different network topologies impact communication and reliability within a network. The star topology proved to be simple to implement but has a single point of failure at the central switch. In contrast, the partial mesh topology provided improved fault tolerance by offering alternative paths for data transmission, although it introduces more complexity. Overall, the activity highlighted the importance of choosing the appropriate topology based on network requirements such as cost, scalability, and reliability.

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